BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type Lahore Board

Note: (Group-I)

Q1. According to endosymbiont hypothesis, the aerobic bacteria developed into :

(A) Ribosome
(B) Lysosome
(C) Mitochondria
(D) Plastids

Q2. Hatching period of chick is :

(A) 15 days
(B) 20 days
(C) 21 days
(D) 25 days

Q3. Nissl's granules are group of:

(A) Mesosomes
(B) Lysosomes
(C) Ribosomes
(D) Chromosomes

Q4. Full cell cycle in yeast cells has length of :

(A) 30 minutes
(B) 60 minutes
(C) 90 minutes
(D) 120 minutes

Q5. Morphological characteristics of chromosomes are collectively called :

(A) Karyotype
(B) Neotype
(C) Holotype
(D) Phenotype

Q6. In 1 gm of ammonia nitrogen requires how much water for excretion :

(A) 50 ml
(B) 100 ml
(C) 200 ml
(D) 500 ml

Q7. Establishment of new forests, where no forests existed before is called

(A) Reforestation
(B) Afforestation
(C) Deforestation
(D) Desertification

Q8. Which one is the type of asexual reproduction?

(A) Apomixes [sic, print for 'Apomixis']
(B) Vernalization
(C) Fertilization
(D) Phototropism

Q9. The enzyme which joins the two pieces of DNA is :

(A) DNA ligase
(B) DNA polymerase
(C) Endo nuclease
(D) Lipase

Q10. Contractile vacuoles are found in :

(A) Plants
(B) Fresh water protozoa
(C) Land animals
(D) Land plants

Q11. DNA synthesis and chromosomal doubling occurs in:

(A) G1-phase
(B) G2-phase
(C) S-phase
(D) S-phase

Q12. The process of moulting is controlled by enzyme called :

(A) Ecdysone
(B) Aldosteron [sic, print for 'Aldosterone']
(C) Oxytocin
(D) Androgen

Q13. Which of the following is vestigial organ of whole [sic, print for 'whale'] :

(A) Gills
(B) Leg bones
(C) Lungs
(D) Pelvis and leg bones

Q14. Locus is :

(A) Part of DNA
(B) Position of gene
(C) Partner of gene
(D) Part of gene

Q15. The abiotic component of an ecosystem is :

(A) Temperature
(B) Producer
(C) Consumer
(D) Decomposer

Q16. Immediately after fertilization, the egg under goes some series of mitotic divisions called :

(A) Morulla [sic, print for 'Morula']
(B) Blastula
(C) Gastrulation
(D) Cleavage

Q17. Mature bone cells are called :

(A) Osteoblast
(B) Osteocytes
(C) Osteoclasts
(D) Chondrocytes

Essay Type

Time Allowed: 2.40 hours | **Maximum Marks:** 68

> Note: Essay Type Group - I Lahore Board

Q2. (Marks 16) Write short answers to any EIGHT (8) questions: (8x2=16)

(i) Differentiate between hydrophytes and mesophytes.

(ii) What are osmoconformers and osmoregulators animals?

(iii) How vasodilation differ from vasoconstriction?

(iv) Define ecdysis or moulting, give its two stages.

(v) Differentiate between troponin and tropomyosin.

(vi) Give two functions of skeletal system.

(vii) Define seed dormancy. Write its two significance.

(viii) Define oviparous and viviparous animals.

(ix) What is profundal zone?

(x) How many biomes are present in the World, name any four of them.

(xi) Differentiate between deforestation and reforestation.

(xii) Define eutrophication, give its one effect upon animal life.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions: (8x2=16)

(i) Differentiate between kinesis and taxes.

(ii) Write the role of progesterone.

(iii) What is Addison's disease?

(iv) What is gene and its locus?

(v) Whati s [sic, print for 'What is'] dihybrid cross?

(vi) Write dorminant [sic, print for 'dominant'] and recessive trait.

(vii) What are three possible ways to get a gene?

(viii) Write a role of Lambda phage as a vector.

(ix) Write any two uses of PCR.

(x) What is Niche, explain according to Charles Eltan?

(xi) What are decomposers?

(xii) Write crustose lichens in xerosere.

Q4. (Marks 12) Write short answers to any SIX (6) questions: (6 x 2 = 12)

(i) Define growth and development.

(ii) Define teratogens. Give two examples.

(iii) Draw structural formula of nucleotide.

(iv) Differentiate between leading strand and lagging strand.

(v) Define transformation. In which bacterium it was discovered?

(vi) What are cancer cells? How cancer cells can be distinguished from normal cells?

(vii) What is meant by non-disjunction? Write its consequences.

(viii) Differentiate between homologous and analogous organs.

(ix) How the oxygen accumulation liberated during photosynthesis changed the environment of earth?

> Note: PART-I [sic, print for 'PART-II']

> Note: Note: Attempt any THREE questions.

Q5. (a) Give the homeostatic roles of liver in the form of a table. (b) Define ecosystem. Discuss its components and their interaction.

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Q6. (a) Write down four phases in the repair process of a fracture. (b) Write a note on genetic code.

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(b) Write a note on genetic code.

Q7. (a) Explain structure and function of forebrain in man. (b) Write a note on algal bloom or eutrophication.

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(b) Write a note on algal bloom or eutrophication.

Q8. (a) Write a note on identical twins and fraternal twins. (b) Write a note on diabetes mellitus.

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Q9. (a) Describe the types of meristems. (b) Explain the evidences of evolution by fossil record and comparative anatomy.

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(b) Explain the evidences of evolution by fossil record and comparative anatomy.

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

> Note: Objective Type (G-II) Lahore Board

> Note: (Group-II; scan note box misprints 'Group-I')

Q1. The stage which may lasts for days, weeks or even years is:

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diplotene

Q2. The follicle cells after release of the egg are modified to form special structure called:

(A) Follicle atresia
(B) Corpus luteum
(C) Uterus
(D) Placenta

Q3. The patients lack a gene that code for trans-membrane carrier of the chloride ions :

(A) Cancer
(B) ADA
(C) SCID
(D) Cystic fibrosis

Q4. The hormone which actively transport water from filtrate in collecting tubules back to kidney is :

(A) Aldosterone
(B) ADH
(C) Testosteron
(D) Oxytocine

Q5. The rain fall less than 25 to 50 cm is found in :

(A) Desert
(B) Grassland
(C) Temperature deciduous forest
(D) Tropical rain forest

Q6. Healing of fracture and repair of the skin wound is example of:

(A) Meiosis
(B) Regeneration
(C) Development
(D) Necrosis

Q7. Which one of the following is a facial bone :

(A) Frontal
(B) Occipital
(C) Vomer
(D) Sternum

Q8. The blood passing through glomerulus is filtered into:

(A) Bowman's capsule
(B) Ureter
(C) Bladder
(D) Urethra

Q9. XYY condition is found in :

(A) Patau
(B) Edward
(C) Turner
(D) Jacobs

Q10. Which of the following is biotic factor :

(A) Topography
(B) Gravity
(C) Soil energy
(D) Decomposers

Q11. If the centrometere [sic, print for 'centromere'] is located in the middle of the chromosome it is called:

(A) Metacentric
(B) Sub metacentric
(C) Telocentric
(D) Acrocentric

Q12. The part of the brain which is best developed in birds:

(A) Cerebellum
(B) Medulla
(C) Hippocampus
(D) Pons

Q13. In Pakistan grassland ecosystem is found in :

(A) Kara Koram
(B) Shogran
(C) Malam Jabba
(D) North Kallat

Q14. In the development of chick the 24 hour embryo is called:

(A) Morulla [sic, print for 'Morula']
(B) Gastrula
(C) Blastula
(D) Neurula

Q15. In plant cell turgor pressure is generated :

(A) Cell wall
(B) Cell membrane
(C) Mitochondria
(D) Vacuole

Q16. The Ginkgo plant is :

(A) Monoecious
(B) Dioecious
(C) Trieciouous [sic, print for 'Triecious']
(D) Polyecious

Q17. The actual remains or traces of organisms that lived in the ancient geological times are called:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

> Note: Essay Type - Group II Subjective: scan at ilmkidunya URL ...-group-ii-subjective-both-medium-101940.aspx shows a paper headed 'Gujranwala Board' (header: Biology | 421-(INTER PART-II) | Paper-II, Time:2.40 hours, Essay Type Group - II, Gujranwala Board), i.e. the WRONG board. Not transcribed per policy; the genuine BISE Lahore Biology Group II Subjective 2021 scan was not found on the source page.